

Inspect Before Modeling

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Python and Table Readiness

Without opening the old lesson, explain one key idea from Python and Table Readiness. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Features, Labels, Targets

Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Interpret the audit counts for Dataset inspection and cards. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Inspect and document independently

A new fictional indoor trip export has (P,4), (Q,None), (Q,None), (R,-1), (S,6), (T,0), with durations in minutes and the same schema. Count raw rows, distinct IDs, missing durations, negative durations and extra ID occurrences. Identify overlap. Draft a short dataset card stating source, row meaning, unit, findings, unchanged version and one excluded use.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A table contains 8 cm, -2 cm, missing and two copies of the same leaf photo. Propose a separate inspection for each issue.

6. Solve the given case

Lengths include 5 cm, blank and -3 cm, plus a duplicate photo row.

Name checks and a deletion risk.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A table has height_cm values 12, 15, -8 and a blank. Identify two issues and a suitable first action.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 5 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Audit the Plant Classifier table and attach a data-quality report before any baseline or model is trained.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A table contains 8 cm, -2 cm, missing and two copies of the same leaf photo. Propose a separate inspection for each issue.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Define the row and source

For Robot practice v1, state what one row is intended to represent, the source, the unit and the meaning of None.

My reasoning and evidence

13. W2 Interpret the audit counts

Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

My reasoning and evidence

14. W3 Reject an unsupported correction

A learner changes None to 0 and -2 to 2, then reports a clean dataset. Explain why this is not a justified repair and what record should be kept.

My reasoning and evidence

15. W4 Inspect and document independently

A new fictional indoor trip export has (P,4), (Q,None), (Q,None), (R,-1), (S,6), (T,0), with durations in minutes and the same schema. Count raw rows, distinct IDs, missing durations, negative durations and

extra ID occurrences. Identify overlap. Draft a short dataset card stating source, row meaning, unit, findings, unchanged version and one excluded use.

My reasoning and evidence

16. W5 State an evidence gap

Someone duplicates each row ten times and claims the export now covers more kinds of trips. Explain what changed and what did not.

My reasoning and evidence

17. Units conversions and data provenance — W1 Retain the source

Name four pieces of information to keep for Log A and its conversion.

My reasoning and evidence

18. Units conversions and data provenance — W2 Convert precisely

Convert 150 seconds and 30 seconds to minutes without rounding to whole numbers.

My reasoning and evidence

19. Units conversions and data provenance — W3 Repair a comparison

A report claims 120 seconds exceeds 3 minutes because $120 > 3$. Correct it.

My reasoning and evidence

20. Units conversions and data provenance — W4 Normalize

independently

Create records for 240 seconds, 1.5 minutes and 60 seconds. Derive minutes while keeping raw data. State the values and their total.

My reasoning and evidence

21. Units conversions and data provenance — W5 Handle an unknown unit

A record contains value 8 and unit ticks with no conversion rule. Should you assume seconds?

My reasoning and evidence

22. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

23. Pictorial case practice: Inspect Before Modeling

Spot a duplicate, an impossible value and a valid boundary before fitting.

Toy heights table: A=3, B=5, B=5, C=-2, D=0 cm. Each plant should have exactly one nonnegative height record.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Inspect Before Modeling

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count rows

Five raw rows represent four distinct IDs; B is duplicated.



2. Check schema

C violates the nonnegative-height rule; D=0 is valid under this contract.



3. Flag before fitting

Flag duplicate B and invalid C. Do not count B twice or silently replace C.

CHANGE ONE THING

The teacher confirms B's second row is an accidental duplicate.

Explain the changed result using the matching lesson.

Which original information should the inspection log preserve? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

24. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

25. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Spot a duplicate, an impossible value and a valid boundary before fitting.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Flag duplicate B and invalid C. Do not count B twice or silently replace C.

1. Flag before fitting
2. Count rows
3. Check schema

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Flag before fitting Flag duplicate B and invalid C. Do not count B twice or silently replace C. Read the picture in order. Flag duplicate B and invalid C. Do not count B twice or silently replace C. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: The teacher confirms B's second row is an accidental duplicate. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. Remove that duplicate with a recorded reason; C still needs correction or explicit exclusion.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Remove that duplicate with a recorded reason; C still needs correction or explicit exclusion. Resolving one defect does not resolve a separate impossible measurement.

Repair a missing building block

Review Python and Table Readiness (ai-4/4). Goal: Specify a table schema. Short practice: Why should the missing value not silently become zero? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Features, Labels, Targets (ai-4/2). Goal: Separate prediction inputs from observed targets. Short practice: Why can a historical table contain a field that is an invalid input? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
